Course Code: B20AM3101

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A) | **R20**

III B.Tech. I Semester MODEL QUESTION PAPER

DESIGN AND ANALYSIS OF ALGORITHMS

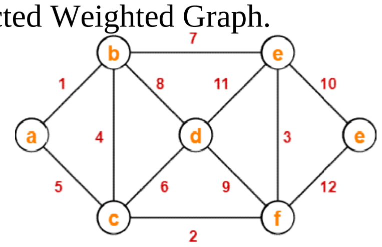
Common to AIML & CIC

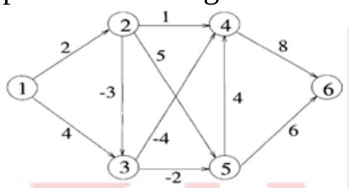
Time: 3 Hrs. | **Max. Marks: 70 M**

Answer **ONE Question** from **EACH UNIT**

All questions carry equal marks

Assume suitable data if necessary

			CO	**KL**	**M**
		UNIT-I			
1.	**a).**	Explain how algorithms performance is analyzed? Describe asymptotic notations	**1**	**2**	**6**
	b).	What is Strassen's matrix multiplication? Explain its time complexity?	**1**	**2**	**8**
		OR			
2.	**a).**	Discuss in detail the Time complexity analysis of Quick sort algorithm with an example.	**1**	**2**	**6**
	b).	Draw the tree of calls of merge sort for the following set. (35, 25,15,10,45, 75, 85, 65, 55, 5, 20, 18)	**1**	**2**	**8**
		UNIT-II			
3.	**a).**	Apply the Greedy algorithm for Job sequencing with deadlines and profits.	**2**	**3**	**6**
	b).	Illustrate how to find the minimum cost spanning tree by using Prim's Algorithm.	**2**	**3**	**8**
		OR			
4.	**a).**	Solve knapsack problem with greedy approach for instances n=3, m=6, profits are (p1, p2, p3) =(1,2,5), weights are(w1,w2,w3)=(2,3,4).	**2**	**3**	**6**
	b).	What is a Minimum cost spanning tree? Compute an efficient data structure for implementation of Krushkal's Algorithm for the connected Weighted Graph.  	**2**	**3**	**8**
		UNIT-III			
5.	**a).**	Construct an optimal travelling sales person tour using Dynamic Programming. 0 10 9 3	**3**	**3**	**7**

<table>
<tr><td></td><td></td><td>5 0 6 2
9 6 0 7
7 3 5 0</td><td></td><td></td><td></td></tr>
<tr><td></td><td>b).</td><td>Apply dynamic programming to obtain optimal binary search tree for the identifier set (a1, a2, a3, a4)=(cin, for, int, while) with (p1, p2, p3, p4)=(1, 4, 2, 1),(q0, q1, q2, q3, q4)=(4, 2, 4, 1, 1) and also write algorithm for its construction.</td><td>3</td><td>3</td><td>7</td></tr>
<tr><td></td><td></td><td>OR</td><td></td><td></td><td></td></tr>
<tr><td>6.</td><td>a).</td><td>Solve 0/1 knapsack problem for the following data using sets method
Let us consider that the capacity of the knapsack is W = 25 and the items are as shown in the following table.
<table>
<tr><th>Item</th><th>A</th><th>B</th><th>C</th><th>D</th></tr>
<tr><td>Profit</td><td>24</td><td>18</td><td>18</td><td>10</td></tr>
<tr><td>Weight</td><td>24</td><td>10</td><td>10</td><td>7</td></tr>
</table></td><td>3</td><td>3</td><td>7</td></tr>
<tr><td></td><td>b).</td><td>Find the shortest path from node 1 to every other node in the graph given below using all pairs shortest algorithm.

</td><td>3</td><td>3</td><td>7</td></tr>
<tr><td></td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td></td><td></td><td>UNIT-IV</td><td></td><td></td><td></td></tr>
<tr><td>7.</td><td>a).</td><td>Relate Hamiltonian cycle with travelling sales person problem and also give the backtracking solution vector that finds all Hamiltonian cycles for any directed or undirected graph.</td><td>4</td><td>3</td><td>8</td></tr>
<tr><td></td><td>b).</td><td>Apply the branch and bound algorithm to generate minimum length tour for the given cost adjacency matrix.
$$\begin{bmatrix} \infty & 18 & 28 & 8 & 9 \\ 13 & \infty & 14 & 2 & 1 \\ 1 & 3 & \infty & 1 & 2 \\ 17 & 4 & 16 & \infty & 1 \\ 14 & 2 & 5 & 16 & \infty \end{bmatrix}$$</td><td>4</td><td>3</td><td>6</td></tr>
<tr><td></td><td></td><td>OR</td><td></td><td></td><td></td></tr>
<tr><td>8.</td><td>a).</td><td>Draw the portion of state space tree generated by recursive backtracking algorithm for sum of subsets problem with an example.</td><td>4</td><td>3</td><td>8</td></tr>
<tr><td></td><td>b).</td><td>Consider the knapsack intance n=4, (p1,p2,p3,p4)=(10,10,12,18),(w1,w2,w3,w4)=(2,4,6,9) and m=15. Solve this 0/1knapsack problem using least cost branch and bound.</td><td>4</td><td>3</td><td>6</td></tr>
<tr><td></td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td></td><td></td><td>UNIT-V</td><td></td><td></td><td></td></tr>
<tr><td>9.</td><td>a).</td><td>Write short notes on</td><td>5</td><td>2</td><td>8</td></tr>
</table>

		i) Classes of NP-hard ii) Classes of NP-complete and Prove that if $NP \neq CO - NP$, then $P \neq NP$			
	b).	Draw the Decision Tree and Find the number of Key Comparison in the worst and average case for: a. The four Element Binary search b. the Three-element basic insertion sort	**5**	**2**	**6**
		OR			
10.	**a).**	Explain P, NP-hard, NP-complete problems and their relationship using pi diagram. Discuss about the Cooks Theorem.	**5**	**2**	**8**
	b).	Explain about the various string matching algorithms.	**5**	**2**	**6**

CO-COURSE OUTCOME KL-KNOWLEDGE LEVEL M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 14 marks

Course Code: B20CI3101

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A) **R20**

III B.Tech. I Semester MODEL QUESTION PAPER

IoT ARCHITECTURE AND ITS PROTOCOLS

For CIC

Time: 3 Hrs. **Max. Marks: 70 M**

Answer **ONE Question** from **EACH UNIT**

All questions carry equal marks

Assume suitable data if necessary

			CO	KL	M
		UNIT-I			
1.	**a).**	Discuss the challenges of IoT.	**1**	**2**	**7**
	b).	Summarize the role of each layer in the IoT World Forum architecture.	**1**	**2**	**7**
		OR			
2.	**a).**	Outline the drivers behind new network Architectures.	**1**	**2**	**7**
	b).	Highlight the key features of OneM2M IoT Architecture.	**1**	**2**	**7**
		UNIT-II			
3.	**a).**	Explain on what basis sensors are classified. Give example for each.	**2**	**2**	**7**
	b).	Draw and explain the high level Zigbee protocol stack.	**2**	**2**	**7**
		OR			
4.	**a).**	Explain on what basis actuators are categorized Give example for each.	**2**	**2**	**7**
	b).	Draw and summarize the layers of LoRaWAN.	**2**	**2**	**7**
		UNIT-III			
5.	**a).**	Summarize the key advantages of Internet Protocol in IoT.	**3**	**2**	**7**
	b).	Differentiate CoAP and MQTT.	**3**	**2**	**7**
		OR			
6.	**a).**	Explain about Optimizing IP for IoT.	**3**	**2**	**7**
	b).	Discuss the core functions of edge analytics.	**3**	**2**	**7**
		UNIT-IV			
7.	**a).**	Explain the Purdue model for control hierarchy.	**4**	**2**	**7**
	b).	Summarize the common challenges in OT security.	**4**	**2**	**7**
		OR			
8.	**a).**	Highlight and discuss the major layers of smart city IoT architecture.	**4**	**2**	**7**
	b).	Draw and explain the smart city traffic architecture.	**4**	**2**	**7**
		UNIT-V			
9.	**a).**	Identify the key challenges and requirements of oil and gas industry.	**5**	**3**	**7**
	b).	Discuss few Use cases of IoT in Oil and gas industry related to oil field	**5**	**3**	**7**

		and connected pipelines.			
		OR			
10.	**a).**	Discuss few use cases of IoT in transportation.	**5**	**3**	**7**
	b).	Identify the major components of the DSRC General Communication IoT Architecture for Transportation.	**5**	**3**	**7**

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

NOTE: Questions can be given as A,B splits or as a single Question for 14 marks

					Course Code: B20CI3102
		SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)			**R20**
		III B.Tech. I Semester MODEL QUESTION PAPER			
		COMPUTER COMMUNICATION AND NETWORKS			
		For CIC			
Time: 3 Hrs.					**Max. Marks: 70 M**
		Answer **ONE Question** from **EACH UNIT**			
		All questions carry equal marks			
		Assume suitable data if necessary			
			CO	**KL**	**M**
		UNIT-I			
1.		Describe the OSI Model and its layers. How does it compare to the TCP/IP protocol suite?	**1**	**2**	**14**
		OR			
2.	**a).**	Explain the effects of transmission impairments on data signals.	**1**	**2**	**4**
	b).	**i.** Using Shannon's theorem, calculate the maximum data rate of a channel with a bandwidth of 3 kHz and a signal-to-noise ratio (SNR) of 20 dB. **ii.** Apply Nyquist theorem to determine the maximum data rate for a noiseless channel with a bandwidth of 4 kHz and 16 signal levels.	**1**	**3**	**10**
		UNIT-II			
3.	**a).**	Discuss the process and importance of cyclic redundancy checks in error detection with an example	**2**	**2**	**7**
	b).	A data block of 1000 bits is sent with a checksum of 16 bits. Calculate the probability of undetected errors if the error probability per bit is 10^3.	**2**	**2**	**7**
		OR			
4.	**a).**	Explain the types of errors that occur in data link layers and how block coding is used for error detection.	**2**	**2**	**7**
	b).	Analyse circuit switching and packet switching methods with suitable example	**2**	**2**	**7**
		UNIT-III			
5.	**a).**	Illustrate the Stop-and-Wait protocol and its significance in data link control.	**3**	**2**	**7**
	b).	Using the Go-Back-N ARQ protocol, calculate the efficiency of the protocol if the window size is 7 and the probability of a frame being damaged is 0.1. Assume no errors in acknowledgment frames.	**3**	**2**	**7**
		OR			
6.	**a).**	Describe the media access control techniques of CSMA/CD and CSMA/CA. How do they differ?	**3**	**2**	**7**
	b).	A network uses FDMA with a bandwidth of 1 MHz divided into 10 equal channels. Calculate the bandwidth of each channel and discuss the	**3**	**2**	**7**

		implications of FDMA on network performance			
		UNIT-IV			
7.	**a).**	A network on the Internet has a subnet mask of 255.255.240.0. What maximum number of hosts it can handle?	**4**	**3**	**7**
	b).	Discuss the services provided by the network layer and the importance of packet switching.	**4**	**2**	**7**
		OR			
8.	**a).**	A block of addresses is granted to a small organization. We know that one of the addresses is 205.16.37.39/28. i. What is the first address in the block? Find the last address for the block c. Find the number of addresses	**4**	**3**	**7**
	b).	Describe the differences between distance vector routing and link-state routing protocols	**4**	**2**	**7**
		UNIT-V			
9.	**a).**	Illustrate the TCP three-way handshake process for connection establishment.	**5**	**2**	**7**
	b).	Explain the function and significance of DNS in the application layer.	**5**	**2**	**7**
		OR			
10.	**a).**	Discuss the role of congestion control in the transport layer and how it is managed.	**5**	**2**	**7**
	b).	Discuss the DNS resolution process and its impact on Internet connectivity.	**5**	**2**	**7**

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

NOTE: Questions can be given as A,B splits or as a single Question for 14 marks

					Course Code: B20CI3103
		SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)			**R20**
		III B.Tech. I Semester MODEL QUESTION PAPER			
		COMPILER DESIGN			
		For CIC			
Time: 3 Hrs.					**Max. Marks: 70 M**
		Answer **ONE Question** from **EACH UNIT**			
		All questions carry equal marks			
		Assume suitable data if necessary			
			CO	**KL**	**M**
		UNIT-I			
1.	**a).**	Construct the output generated by different phases of compiler for the following A=B+C*20	**1**	**3**	**7**
	b).	Explain about Recognition of tokens.	**1**	**2**	**7**
		OR			
2.	**a).**	Construct the parse tree for the following statement While(A<B) Do A=A+1	**1**	**3**	**7**
	b).	Explain the Role of Lexical analyzer.	**1**	**2**	**7**
		UNIT-II			
3.	**a).**	Differentiate top-down and Bottom-up parsing.	**2**	**3**	**7**
	b).	Apply Shift-Reduce parser to determine whether the string id+id*id is generated from the following grammar or not E□E+T/T T□T*F/F F□(E)/id	**2**	**3**	**7**
		OR			
4.	**a).**	Determine FIRST & FOLLOW for the following grammar S□AB A□aA/€ B□bB/b	**2**	**3**	**7**
	b).	Construct LR (1) sets of items for the following CFG. S□AA A□aA/b	**2**	**3**	**7**
		UNIT-III			
5.	**a).**	Describe how SDTs can be used to generate intermediate code from a parse tree.	**3**	**2**	**7**
	b).	Convert the following arithmetic expression into three-address code: (a + b) * (c - d)	**3**	**3**	**7**

		OR			
6.	**a).**	Translate the expression a = (b * -c) + (b * -c) into Quadruples, triples and indirect triples.	**3**	**3**	**7**
	b).	Differentiate L-attributed SDT and S-attributed SDT	**3**	**3**	**7**
		UNIT-IV			
7.	**a).**	Explain about Heap Management.	**4**	**2**	**7**
	b).	Explain about data-flow analysis on basic blocks and reaching definitions.	**4**	**2**	**7**
		OR			
8.	**a).**	Construct DAG for the arithmetic expression a + a*(b-c) + (b-c) *d	**4**	**3**	**7**
	b).	Explain about the principal sources of optimization.	**4**	**2**	**7**
		UNIT-V			
9.	**a).**	Generate target code from a sequence of three address statements using a simple code generator algorithm.	**5**	**3**	**7**
	b).	Discuss about peephole optimization techniques.	**5**	**2**	**7**
		OR			
10.	**a).**	Generate target code from a sequence of three address statements using code generation from DAG.	**5**	**3**	**7**
	b).	Discuss issues in the Design of a Code Generator.	**5**	**2**	**7**

CO-COURSE OUTCOME KL-KNOWLEDGE LEVEL M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 14 marks

Course Code: B20CI3104					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R20
III B.Tech. I Semester MODEL QUESTION PAPER					
SOFTWARE ENGINEERING					
For CIC					
Time: 3 Hrs.					Max. Marks: 70 M
Answer **ONE Question** from **EACH UNIT**					
All questions carry equal marks					
Assume suitable data if necessary					
			CO	**KL**	**M**
		UNIT-I			
1.	**a).**	Explain about Nature of the software.	**1**	**2**	**7**
	b).	Explain about Evolutionary process models.	**1**	**2**	**7**
		OR			
2.	**a).**	Explain about Software Myths.	**1**	**2**	**7**
	b).	Explain about Agile Process.	**1**	**2**	**7**
		UNIT-II			
3.	**a).**	Explain the concept of use cases with the help of an example.	**2**	**2**	**7**
	b).	Explain the process of requirements engineering.	**2**	**2**	**7**
		OR			
4.	**a).**	State and explain various aspects in the requirements validation process.	**2**	**2**	**7**
	b).	Elaborate the main focus of requirement analysis.	**2**	**2**	**7**
		UNIT-III			
5.	**a).**	Explain about UML models that supplement use cases.	**3**	**2**	**7**
	b).	What is Class based Modeling? Explain Elements of Class based Modeling.	**3**	**2**	**7**
		OR			
6.	**a).**	Explain the steps to create a behavior model for a System.	**3**	**2**	**7**
	b).	Explain about the requirements patterns with an example.	**3**	**2**	**7**
		UNIT-IV			
7.	**a).**	Explain about different architecture Styles.	**4**	**2**	**8**
	b).	Explain Web Apps Interface Design.	**4**	**2**	**6**
		OR			
8.	**a).**	Explain the process of Designing Class-Based Components for WebApps.	**4**	**2**	**8**
	b).	Explain the Golden Rules for User Interface Design.	**4**	**2**	**6**
		UNIT-V			

9.	**a).**	Explain the testing strategies for conventional software.	**5**	**2**	**7**
	b).	What is Debugging? Explain about the art of Debugging.	**5**	**2**	**7**
		OR			
10.	**a).**	Explain the testing strategies for Object Oriented software.	**5**	**2**	**7**
	b).	Explain basic path testing with one example.	**5**	**3**	**7**

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

NOTE: Questions can be given as A,B splits or as a single Question for 14 marks

Course Code: B20CI3105					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					**R20**
III B.Tech. I Semester MODEL QUESTION PAPER					
ADVANCED COMPUTER ARCHITECTURE					
For CIC					
Time: 3 Hrs.					**Max. Marks: 70 M**
Answer **ONE Question** from **EACH UNIT**					
All questions carry equal marks					
Assume suitable data if necessary					
			CO	**KL**	**M**
		UNIT-I			
1.	**a).**	Explain about processes and threads in languages.	**1**	**2**	**7**
	b).	Illustrate the concept of Classification of Parallelism.	**1**	**2**	**7**
		OR			
2.	**a).**	Describe in detail about Flynn's Classification.	**1**	**2**	**7**
	b).	Describe the Utilized Parallelism.	**1**	**2**	**7**
		UNIT-II			
3.	**a).**	Describe the concept of Evolution of ILP	**2**	**2**	**7**
	b).	Explain about Pipelined Instruction Processing	**2**	**2**	**7**
		OR			
4.	**a).**	Illustrate Control and Resource dependencies	**2**	**2**	**7**
	b).	Explain about principles and general structure of pipelines	**2**	**2**	**7**
		UNIT-III			
5.	**a).**	Illustrate Proposed and Commercial VLIW Architectures	**3**	**2**	**7**
	b).	Describe the parallel decoding	**3**	**2**	**7**
		OR			
6.	**a).**	Describe in detail about superscalar instruction issue	**3**	**2**	**7**
	b).	Explain about preserving sequential consistency of instruction execution.	**3**	**2**	**7**
		UNIT-IV			
7.	**a).**	Determine Basic approaches to branch handling	**4**	**3**	**7**
	b).	Compute the branch processing	**4**	**3**	**7**
		OR			
8.	**a).**	Determine the delayed branching	**4**	**3**	**7**
	b).	Compute the multiway branching guarded execution	**4**	**3**	**7**
		UNIT-V			
9.	**a).**	Summarize the concept of Near neighbours	**5**	**2**	**7**

	b).	Describe in detail about Reconfigurable Networks	**5**	**2**	**7**
		OR			
10.	**a).**	Compare the structures of hypercube and pyramid	**5**	**2**	**7**
	b).	Explain about Alternative Architectural Classes	**5**	**2**	**7**

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

NOTE: Questions can be given as A,B splits or as a single Question for 14 marks

Course Code: **B20CI3106**

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)	**R20**

III B.Tech. I Semester MODEL QUESTION PAPER

COMPUTER GRAPHICS

For CIC

Time: 3 Hrs. **Max. Marks: 70 M**

Answer **ONE Question** from **EACH UNIT**

All questions carry equal marks

Assume suitable data if necessary

			CO	KL	M
		UNIT-I			
1.	**a).**	List and explain applications of Computer Graphics	**1**	**2**	**7**
	b).	Explain Raster-Scan systems with figures.	**1**	**2**	**7**
		OR			
2.	**a).**	Explain basic components of a computer graphics system	**1**	**2**	**7**
	b).	Illustrate generation of images on the screen in a random-scan system	**1**	**2**	**7**
		UNIT-II			
3.	**a).**	Apply simple DDA-Line drawing algorithm steps for an example line.	**2**	**3**	**7**
	b).	Determine transformation matrix and the coordinates of a square ((0,0), (100,0), (100,100), (0,100)) after scaling it by a factor (0.5,1).	**2**	**3**	**7**
		OR			
4.	**a).**	Demonstrate how basic 2D geometric transformations are performed on objects.	**2**	**3**	**7**
	b).	Apply Bresenham's line algorithm for scan converting a line form (10,10) to (20,16).	**2**	**3**	**7**
		UNIT-III			
5.	**a).**	Demonstrate the approach used by the Nicholl-Lee-Nicholl algorithm to clip lines.	**3**	**3**	**7**
	b).	Apply Liang-Barsky line clipping algorithm to clip an example line against a square window ((0,0), (100,0), (100,100), (0,100)).	**3**	**3**	**7**
		OR			
6.	**a).**	Determine the transformation matrix for Window-to-Viewport coordinate transformation	**3**	**3**	**7**
	b).	Demonstrate Cohen-Sutherland line Clipping Algorithm with diagrams.	**3**	**3**	**7**
		UNIT-IV			
7.	**a).**	Illustrate wire frame model of Solid modelling	**4**	**2**	**7**
	b).	Describe parametric representation for Bezier curves and surfaces.	**4**	**2**	**7**
		OR			

8.	**a).**	Describe B-spline curves with figures.	**4**	**2**	**7**
	b).	Illustrate perspective projections	**4**	**2**	**7**
		UNIT-V			
9.	**a).**	Demonstrate basic Three-Dimensional Translation and Scaling.	**5**	**3**	**7**
	b).	Illustrate motion specification in the context of computer animation?	**6**	**2**	**7**
		OR			
10.	**a).**	Determine 3D transformation matrices for rotation about X-axis, Y-axis and Z-axis.	**5**	**3**	**7**
	b).	Compare raster animation and vector animation?	**6**	**2**	**7**

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

NOTE: Questions can be given as A,B splits or as a single Question for 14 marks

Course Code: B20CI3201

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)	**R20**

III B.Tech. II Semester MODEL QUESTION PAPER

CRYPTOGRAPHYANDNETWORK SECURITY

For CIC

Time: 3 Hrs. **Max. Marks: 70 M**

Answer **ONE Question** from **EACH UNIT**

All questions carry equal marks

Assume suitable data if necessary

			CO	**KL**	**M**
		UNIT-I			
1.	**a).**	Differentiate Active attacks and Passive attacks.	**1**	**2**	**7**
	b).	Explain Traditional Block Cipher Structure	**1**	**2**	**7**
		OR			
2.	**a).**	Illustrate TCP session hijacking, UDP Session Hijacking.	**1**	**2**	**7**
	b).	Explain Block cipher design principles.	**1**	**2**	**7**
		UNIT-II			
3.	**a).**	Given the system of congruences: x≡2(mod3) x≡3(mod5) x≡2(mod7) Use the Chinese Remainder Theorem to find the smallest positive integer that satisfies all three congruences.	**2**	**3**	**7**
	b).	ExplainthestructureofAESalgorithmwithneatdiagramanddescribethesteps in AES encryption.	**2**	**2**	**7**
		OR			
4.	**a).**	Let p=11p = 11, a prime number. Compute 3^{20}mod11 using Fermat's Little Theorem.	**2**	**3**	**7**
	b).	Explain Block Cipher modes of operations.	**2**	**2**	**7**
		UNIT-III			
5.	**a).**	Perform Encryption and Decryption using RSA algorithm for p=17,q=11,e=7,M=88. Demonstrate digital signature algorithm with neat diagram and explain how to sign and verify using DSS algorithm.	**3**	**3**	**7**
	b).	Differentiate between HMAC and CMAC	**4**	**3**	**7**
		OR			
6.	**a).**	Illustrate various steps ofSHA-512in detail with neat diagram.	**4**	**2**	**7**
	b).	FindthesecretkeysharedbetweenUserAandUserBusingDiffie HellmanKeyexchange algorithm for the following:q=97,a=5, the private keys XA= 36,XB= 58.	**3**	**3**	**7**

		UNIT-IV			
7.	**a).**	Describe IP sec architecture with neat diagram.	**5**	**2**	**7**
	b).	Illustrate the services provided by PGP with neat diagram.	**5**	**2**	**7**
		OR			
8.	**a).**	Discuss in detail about user Authentication using Kerberos protocol.	**5**	**2**	**7**
	b).	Explain Key Management using OAKLEY, ISAKMP	**5**	**2**	**7**
		UNIT-V			
9.	**a).**	Describe Secure Socket Layer (SSL).	**6**	**2**	**7**
	b).	Explain about different types of firewalls.	**6**	**2**	**7**
		OR			
10.	**a).**	Illustrate how fir walls are configured.	**6**	**2**	**7**
	b).	Describe the requirements of Web security	**6**	**2**	**7**

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

NOTE: Questions can be given as A,B splits or as a single Question for 14 marks

Course Code: B20CI3202

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A) | **R20**

III B.Tech. II Semester MODEL QUESTION PAPER

INTRODUCTION TO CYBER SECURITY

For CIC

Time: 3 Hrs. **Max. Marks: 70 M**

Answer **ONE Question** from **EACH UNIT**

All questions carry equal marks

Assume suitable data if necessary

			CO	KL	M
		UNIT-I			
1.	**a).**	Categorize various types of threats and give examples for each.	**1**	**2**	**7**
	b).	Identify any seven application attacks and define.	**1**	**2**	**7**
		OR			
2.	**a).**	Discuss methods in securing Wireless LANs (WLAN).	**1**	**2**	**7**
	b).	Explain ICMP attacks with a neat diagram.	**1**	**2**	**7**
		UNIT-II			
3.	**a).**	Explain the windows boot process with a neat flow chart.	**2**	**2**	**7**
	b).	Differentiate hardware based and software based Cyber security countermeasures.	**2**	**3**	**7**
		OR			
4.	**a).**	Explain various tools used in Network based malware protection.	**2**	**2**	**7**
	b).	Differentiate business policy and security policy and give examples for each.	**2**	**3**	**7**
		UNIT-III			
5.	**a).**	Discuss various access control methods.	**3**	**2**	**7**
	b).	Highlight the differences between TACACS+ and RADIUS for AAA.	**3**	**2**	**7**
		OR			
6.	**a).**	Explain about Zone Based Policy Firewalls (ZPF).	**3**	**2**	**7**
	b).	Interpret named and extended ACLs with examples for each.	**3**	**2**	**7**
		UNIT-IV			
7.	**a).**	Identify and discuss the technologies that impact security monitoring protocols.	**4**	**2**	**7**
	b).	Summarize various network security testing tools.	**4**	**2**	**7**
		OR			
8.	**a).**	Categorize the CIS critical security controls.	**4**	**2**	**7**
	b).	Identify various types of security policies.	**4**	**2**	**7**
		UNIT-V			

9.	**a).**	Discuss about various threat intelligence sources.	**5**	**2**	**7**
	b).	Explain the Diamond Model of Intrusion Analysis.	**5**	**2**	**7**
		OR			
10.	**a).**	Discuss about various threat intelligence services.	**5**	**2**	**7**
	b).	Explain the Cyber Kill Chain.	**5**	**2**	**7**

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

NOTE: Questions can be given as A,B splits or as a single Question for 14 marks

					Course Code: B20CI3203
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					**R20**
III B.Tech. II Semester MODEL QUESTION PAPER					
BLOCKCHAIN TECHNOLOGIES					
CIC					
Time: 3 Hrs.					**Max. Marks: 70 M**
Answer **ONE Question** from **EACH UNIT**					
All questions carry equal marks					
Assume suitable data if necessary					
			CO	**KL**	**M**
		UNIT-I			
1.	**a).**	Explain the benefits and limitations of blockchain technology.	**1**	**2**	**6**
	b).	Discuss the various stages in Block chain evolution.	**1**	**2**	**8**
		OR			
2.	**a).**	Explain about the block chain characteristics in detail.	**1**	**2**	**8**
	b).	Differentiate centralized applications and Decentralized applications.	**1**	**2**	**6**
		UNIT-II			
3.	**a).**	Explain the Merkle-tree structure in Block chain.	**2**	**2**	**6**
	b).	Explain the Consensus algorithm in detail	**2**	**2**	**8**
		OR			
4.	**a).**	Discuss about the types of Block chain nodes and the risks associated with block chain solutions.	**2**	**2**	**6**
	b).	Explain in detail the Life cycle of block chain transaction.	**2**	**2**	**8**
		UNIT-III			
5.		Explain the process of designing a block chain applications using Ethereum with a suitable example.	**3**	**3**	**14**
		OR			
6.		Explain the process of designing a block chain applications using Hyper ledger Fabric Architecture with a suitable example.	**3**	**3**	**14**
		UNIT-IV			
7.	**a).**	Explain the smart contract programming with an example	**4**	**3**	**8**
	b).	Explain the Truffle framework and Ganache in Blockchain.	**4**	**2**	**6**
		OR			
8.	**a).**	Explain Remix IDE for Ethereum smart contract development.	**4**	**2**	**7**
	b).	Explain Open Zeppelin Contracts with one example.	**4**	**3**	**7**
		UNIT-V			
9.	**a).**	Explain the Hyper ledger Fabric Transaction Flow in detail.	**5**	**2**	**8**

	b).	Explain Invoking Chain code functions using client application.	**5**	**2**	**6**
		OR			
10.	**a).**	Explain the inter Planetary File System (IPFS) in detail.	**5**	**2**	**8**
	b).	Discuss about the Zero-Knowledge Proofs in Block chain.	**5**	**2**	**6**

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

Course Code: B20CI3204

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A) **R20**

III B.Tech. II Semester MODEL QUESTION PAPER

DATA WAREHOUSING AND DATA MINING

For CIC

Time: 3 Hrs. **Max. Marks: 70 M**

Answer **ONE Question** from **EACH UNIT**

All questions carry equal marks

Assume suitable data if necessary

<table>
<tr><th></th><th></th><th></th><th>CO</th><th>KL</th><th>M</th></tr>
<tr><td></td><td></td><td>UNIT-I</td><td></td><td></td><td></td></tr>
<tr><td>1.</td><td>a).</td><td>Discuss about Multidimensional data model in Data Warehouse?</td><td>1</td><td>2</td><td>7</td></tr>
<tr><td></td><td>b).</td><td>Explain Data warehouse Design and its implementation</td><td>1</td><td>2</td><td>7</td></tr>
<tr><td></td><td></td><td>OR</td><td></td><td></td><td></td></tr>
<tr><td>2.</td><td>a).</td><td>Define data mining? Explain the process of Knowledge Discovery(KDD)?</td><td>1</td><td>2</td><td>7</td></tr>
<tr><td></td><td>b).</td><td>Explain different types of data in Data Mining?</td><td>1</td><td>2</td><td>7</td></tr>
<tr><td></td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td></td><td></td><td>UNIT-II</td><td></td><td></td><td></td></tr>
<tr><td>3.</td><td>a).</td><td>What is data preprocessing? Explain data cleaning in detail?</td><td>2</td><td>2</td><td>7</td></tr>
<tr><td></td><td>b).</td><td>Explain about data transformation strategies?
Use these methods to normalize the following group of data:
200,300,400,600,1000
i. min-max normalization by setting min = 0 and max = 1
ii. z-score normalization.
ii. z-score normalization using the mean absolute deviation instead of standard deviation.</td><td>2</td><td>3</td><td>7</td></tr>
<tr><td></td><td></td><td>OR</td><td></td><td></td><td></td></tr>
<tr><td>4.</td><td>a).</td><td>Discuss about frequent item set mining? A data base has five transactions. Letmin_supD60%andmin_conf80%. Apply Apriori algorithm to find all frequent item sets.
<table>
<tr><th>TID</th><th>Items bought</th></tr>
<tr><td>T100</td><td>{M,O,N,K,E,Y}</td></tr>
<tr><td>T200</td><td>{D,O,N,K,E,Y}</td></tr>
<tr><td>T300</td><td>{M,A,K,E}</td></tr>
<tr><td>T400</td><td>{M,U,C,K,Y}</td></tr>
<tr><td>T500</td><td>{C,O,O,K,I,E}</td></tr>
</table></td><td>2</td><td>3</td><td>7</td></tr>
<tr><td></td><td>b).</td><td>Demonstrate FP Growth algorithm with an example.</td><td>2</td><td>2</td><td>7</td></tr>
<tr><td></td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td></td><td></td><td>UNIT-III</td><td></td><td></td><td></td></tr>
<tr><td>5.</td><td>a).</td><td>Apply Naïve Bayes algorithm on the following dataset and classify the</td><td>3</td><td>3</td><td>7</td></tr>
</table>

		following tuple: *(Outlook=Sunny, Temperature=Cool, Humidity=High, Wind=Strong)*			

	Predictors				Response
	Outlook	Temperature	Humidity	Wind	Class Play=Yes Play=No
Day1	Sunny	Hot	High	Weak	No
Day2	Sunny	Hot	High	Strong	No
Day3	Overcast	Hot	High	Weak	Yes
Day4	Rain	Mild	High	Weak	Yes
Day5	Rain	Cool	Normal	Weak	Yes
Day6	Rain	Cool	Normal	Strong	No
Day7	Overcast	Cool	Normal	Strong	Yes
Day8	Sunny	Mild	High	Weak	No
Day9	Sunny	Cool	Normal	Weak	Yes
Day10	Rain	Mild	Normal	Weak	Yes
Day11	Sunny	Mild	Normal	Strong	Yes
Day12	Overcast	Mild	High	Strong	Yes
Day13	Overcast	Hot	Normal	Weak	Yes
Day14	Rain	Mild	High	Strong	No

	b).	Illustrate cross-validation is in the context of machine learning and describe how K-Fold Cross-Validation works.	**4**	**2**	**7**
		OR			
6.	**a).**	Explain the causes of model over fitting	**4**	**2**	**7**
	b).	Apply Back propagation algorithm for neural network-based Classification of data with an example?	**3**	**3**	**7**
		UNIT-IV			
7.	**a).**	Consider that the data mining task is to cluster the following seven pointsP1,P2,P3,P4,P5,P6,P7intotwoclusters.P1(1,1),P2(2,2),P3(3,4),P4(5,7), P5(3,5),P6(4,5) andP7(4,6). The distance function is Euclidean distance. Apply K-means algorithm with two iterations to form two clusters by taking the initial cluster centres as points P1 andP4?	**5**	**3**	**7**
	b).	Compare partitional and hierarchical clustering algorithms?	**5**	**2**	**7**
		OR			
8.	**a).**	Demonstrate DBSCAN algorithm with the help of an example?	**5**	**3**	**7**
	b).	Illustrate strengths and weakness of K-Mediods algorithm.	**5**	**2**	**7**
		UNIT-V			
9.	**a).**	Explain Characteristics of anomaly detection problem	**6**	**2**	**7**
	b).	Illustrate Statistical approaches, Proximity based approaches for anomaly detection.	**6**	**2**	**7**
		OR			
10.	**a).**	Explain Clustering based approaches for anomaly detection.	**6**	**2**	**7**
	b).	Discuss the Evaluation process of Anomaly Detection.	**6**	**2**	**7**

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

NOTE: Questions can be given as A,B splits or as a single Question for 14 marks

					Course Code: B20CI3205
		SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)			**R20**
		III B.Tech. II Semester MODEL QUESTION PAPER			
		OBJECT ORIENTED ANALYSIS & DESIGN USING UML			
		For CIC			
Time: 3 Hrs.					**Max. Marks: 70 M**
		Answer **ONE Question** from **EACH UNIT**			
		All questions carry equal marks			
		Assume suitable data if necessary			
			CO	**KL**	**M**
		UNIT-I			
1.	**a).**	Explain the attributes of complex system	**1**	**2**	**7**
	b).	Explain the importance of model building	**1**	**2**	**7**
		OR			
2.	**a).**	Explain Algorithmic verses Object Oriented Decomposition	**1**	**2**	**7**
	b).	Explain the elements of Software Design Methodologies	**1**	**2**	**7**
		UNIT-II			
3.	**a).**	Explain modeling simple collaborations with an example	**2**	**2**	**7**
	b).	Explain modeling static types and dynamic types	**2**	**2**	**7**
		OR			
4.	**a).**	Explain the special properties of attributes and operations	**2**	**2**	**7**
	b).	Compare composition and aggregation relationships	**2**	**2**	**7**
		UNIT-III			
5.	**a).**	Explain principles of modeling.	**3**	**2**	**7**
	b).	Explain common techniques for modeling classes and relationships	**3**	**2**	**7**
		OR			
6.	**a).**	Explain common mechanisms in UML	**3**	**2**	**7**
	b).	Explain class diagrams with an example	**3**	**2**	**7**
		UNIT-IV			
7.	**a).**	Explain modeling flows of control by time ordering with an example	**4**	**2**	**7**
	b).	Explain modeling a workflow with suitable swimlane diagram.	**4**	**2**	**7**
		OR			
8.	**a).**	Explain modeling the context of a system with an example use case diagram	**4**	**2**	**7**
	b).	Explain modeling an operation with an example activity diagram	**4**	**2**	**7**
		UNIT-V			
9.	**a).**	Explain the kinds of events.	**5**	**2**	**7**

	b).	Explain modeling the distribution of component with an example.	**5**	**2**	**7**
		OR			
10.	**a).**	Explain modeling the lifetime of an object with an example.	**5**	**2**	**7**
	b).	Explain communication and synchronization when objects collaborate.	**5**	**2**	**7**

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

NOTE: Questions can be given as A,B splits or as a single Question for 14 marks

Course Code: **B20CI3206**

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)	**R20**

III B.Tech. II Semester MODEL QUESTION PAPER

NoSQL DATABASES

For CIC

Time: 3 Hrs. **Max. Marks: 70 M**

Answer **ONE Question** from **EACH UNIT**

All questions carry equal marks

Assume suitable data if necessary

			CO	KL	M
		UNIT-I			
1.	**a).**	Explain the emergence of NoSQL.	**1**	**2**	**7**
	b).	Explain about Location Preference Stores.	**1**	**2**	**7**
		OR			
2.	**a).**	Explain Sorted Ordered Column Oriented Data Stores.	**1**	**2**	**7**
	b).	Discuss about Key value Stores.	**1**	**2**	**7**
		UNIT-II			
3.		Explain about language bindings for NoSQL Data Stores with proper examples.	**2**	**3**	**14**
		OR			
4.		Explain CRUD operation in NoSQL with suitable examples.	**2**	**3**	**14**
		UNIT-III			
5.	**a).**	Explain Hbase Distributed StorageArchitecture.	**3**	**2**	**7**
	b).	Discuss about Key/Value Stores in Memcached and Redis.	**3**	**2**	**7**
		OR			
6.	**a).**	Explain Document Store Internals.	**3**	**2**	**7**
	b).	Explain Consistent Non-Relational Databases.	**3**	**2**	**7**
		UNIT-IV			
7.	**a).**	Explain accessing Data from Column-Oriented DatabasesHbase with proper example.	**4**	**3**	**7**
	b).	Explain Schema evolution In Column-Oriented Databases with example.	**4**	**3**	**7**
		OR			
8.	**a).**	Explain Querying Redis Data Stores with example.	**4**	**3**	**7**
	b).	Explain Data Evolution In Key/Value Stores with example	**4**	**3**	**7**
		UNIT-V			
9.	**a).**	Explain Indexing and Ordering in Mongodb.	**5**	**2**	**7**

	b).	Explain Indexing and Ordering in Couchdb.	**5**	**2**	**7**
		OR			
10.	**a).**	Explain Creating and Using Indexes in Mongodb.	**5**	**2**	**7**
	b).	Explain Indexing InApache Cassandra.	**5**	**2**	**7**

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

NOTE: Questions can be given as A,B splits or as a single Question for 14 marks